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PCN175

THE IMPACT OF A FAST-TRACK COLORECTAL PROGRAM ON INCREASING SURGICAL TREATMENT COVERAGE IN THE PUBLIC HEALTHCARE SYSTEM: SIMULATION ANALYSIS FOR SÃO PAULO STATE

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INTRODUCTION: Colorectal cancer has the second most frequent incidence among men and women in São Paulo state, it was expected 11.560 new cases for 2014. Most of cases require surgical treatment and nowadays the public health care system covers only 54% of the demand. Raise coverage may occur without increase the health care network. A Fast Track program that brings efficiency and consists in a set of perioperative actions that generates less surgical stress, less inflammatory response, promoting a faster recovery and lower complications rate, when compared with standard treatments could help to achieve this goal. Several studies show that it's possible to reduce the hospital stay in 2.5 days with the adoption of a Fast track program, without increasing mortality and readmission rates. **OBJECTIVES:** Evaluate the potential growth coverage in the existing demand by simulating the deployment of a Fast Track program in key public institutions in São Paulo state. **METHODS:** Hospitals with the greatest colorectal cancer surgical volume were selected. Ten services correspond to 50% of the surgical volume in colorectal cancer. Based on the results obtained in the literature and the current hospital stay of each hospital, it was simulated the deployment of a Fast Track program and measured the productivity, considering the same bed infrastructure. **RESULTS:** During the simulation, 9 of 10 hospitals would increase productivity, resulting in a potential growth of 30% in the number of surgeries, from 1,390 to 1,804 procedures. Thus, the adoption of a Fast Track program in 10 institutions would increase the current demand coverage from 54% to 71%. The average length of stay would decrease from 8.7 to 6.7 days. **CONCLUSIONS:** Fast Track program could increase the productivity and efficiency of the colorectal cancer surgical care without raises the health care network, contributing to increase the public health care coverage.

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EXAMINING RACIAL DISPARITIES IN INVASIVE BREAST CANCER AMONG YOUNGER WOMEN: ANALYSIS USING MULTIPLE MEASURES OF POPULATION HEALTH

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OBJECTIVES: Few studies have examined racial disparities in invasive breast cancer among women aged 18-44 years in the United States. This study estimates the magnitude of racial disparities in breast cancer among younger women using five measures of population health: life expectancy (LE), expected years of life lost (EYLL), cumulative incidence rate (CIR), and incidence and mortality rate ratios (IRR and MRR). **METHODS:** The LE and EYLL were estimated for black and white women from a Surveillance, Epidemiology, and End Results cohort of 37,857 women ages 18-44 years diagnosed with invasive breast cancer, 2000-2011. The survival function for an age- and sex-matched reference population was generated using the Monte Carlo method from the life table of the general population. Lifetime survival of the breast cancer patients (up to 60 years) were obtained using linear extrapolation of a logit-transformed curve of the survival ratio. The CIR, IRR and MRR were calculated using 2007-2011 breast cancer incidence and mortality rates from the Centers for Disease Control and Prevention's National Program of Cancer Registries. **RESULTS:** The estimated LE for younger black women ranged from 29.8 for localized stage to 5.3 years for distant stage, and 33.1 to 8.7 years for white women. The estimated EYLL was 17.1 years for black and 13.6 years for white women. The CIR18-44 indicated a 1 in 85 probability for black women compared to a 1 in 92 probability for white women of being diagnosed with breast cancer by age 45. The estimated age-adjusted breast cancer IRR for black-to-white women was 1.06 and the corresponding MRR was 1.88. **CONCLUSIONS:** The persistent breast cancer disparities between younger black compared to white women highlight the need for continued public health efforts to understand etiology, promote primary prevention, and improve access to quality healthcare.

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UTILIZATION OF THE 21-GENE RS TEST AMONG MEDICARE BENEFICIARIES WITH BREAST CANCER IN 2011

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OBJECTIVES: The numbers of genomic tests and indications for testing are rapidly expanding. By identifying patients' response to treatment, genomic risk stratification tools can improve health outcomes, while decreasing costs and toxicities of ineffective treatments. The 21-gene recurrence score test predicts 10-year risk of recurrence and chemotherapy benefit in early-stage, estrogen receptor (ER) positive breast cancer patients. We evaluated utilization of this test among Medicare beneficiaries and analyzed clinical, demographic, and regional factors that predict testing. **METHODS:** Using 2011 Medicare claims data, we conducted a retrospective cohort study of fee-for-service beneficiaries diagnosed with breast cancer who had undergone tissue collection. Independent variables expected to predict testing were patient age, gender, race, Medicaid status, clinical characteristics, and the hospital referral region (HRR). **RESULTS:** Among the 114,624 beneficiaries analyzed, 11,934 (12%) had a claim for the 21-gene test. HRR was the strongest predictor of testing, odds ratios ranging from 0.21 (CI 0.07 - 0.62) for Rochester, Minnesota HRR to 3.61 (CI 1.66 - 7.87) for Slidell, Louisiana. ER-positive patients had an OR of 2.58, CI 2.48-2.69. White patients were more likely to be tested than minorities (OR 1.42, CI 1.33-1.51). Medicaid patients were less likely to be tested (OR 0.75, CI 0.71-0.80). For every

tenth point increase in the weighted average risk score, there was a 5% decrease in likelihood of testing. The likelihood decreased with beneficiaries' age, by 4% per year (OR 0.96, CI 0.96-0.96). **CONCLUSIONS:** Despite widespread implementation of the 21-gene test, some minorities and Medicaid patients have less access to testing. We also identified patients for whom the appropriateness of testing was questionable due to comorbid medical conditions or advanced age. Our data illustrate the need to improve coding to ensure that reimbursement of genomic tests is tied to appropriate use and to discourage overutilization.

PCN178

TOWARDS AN EVIDENCE THRESHOLD AT PCODR

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OBJECTIVES: To determine the quality of evidence required by the Pan-Canadian Oncology Drug Review in order to support a positive or conditional recommendation. **METHODS:** All 64 pCODR reviews from inception to December 31, 2015 were searched for relevant content. Factors of interest included: reliance on abstract-level data as primary clinical information; reliance on early-terminated trials; reliance on non-randomized information; reliance on unblinded trials; and the use of network meta-analyses to establish comparative efficacy. **RESULTS:** The use of abstract-level data appears to be significantly predictive of a negative recommendation, while the other factors have yet to demonstrate significant effects on the quality of recommendations by pCODR. **CONCLUSIONS:** Many factors contribute to success at pCODR, including the availability of other treatment options, (i.e. unmet need), and the burden of illness. The quality of the clinical evidence in support of reimbursement is only one factor among several, though it has clearly been shown to be the most important. Within the domain of clinical evidence, we have determined that certain limitations on the quality of the available evidence have demonstrated a meaningful effect on the likelihood of a positive or conditional recommendation. Manufacturers would therefore be advised to consider these conclusions when determining the advisability of submitting earlier in their clinical development program. The implications of a negative recommendation may substantially outweigh the benefits of earlier access to markets.

PCN179

ECONOMIC OUTCOMES ASSOCIATED WITH TUMOR HISTOLOGY AMONG ELDERLY PATIENTS WITH METASTATIC NON-SMALL CELL LUNG CANCER

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OBJECTIVES: Treatments for metastatic non-small cell lung cancer (mNSCLC) vary depending on histologic subtype and may have differential impact on economic outcomes. We therefore examined disease-related healthcare and cancer-directed treatment costs by histologic subtype among patients with mNSCLC. **METHODS:** We identified patients aged ≥65 years diagnosed with mNSCLC between 2000 and 2011 using the SEER-Medicare linked database. Patients with initial diagnosis at earlier-stage disease were required to have ≥1 subsequent diagnosis of secondary metastases (ICD-9-CM codes: 196.xx-198.xx). The first mNSCLC diagnosis date defined the index date; patients were followed from index until death or database end (12/31/2012). Patients were stratified by squamous versus nonsquamous histology. Disease-related costs consisted of Medicare-reimbursed costs on healthcare claims for inpatient admissions, ambulatory care visits, and hospice care with a cancer-related diagnosis code, as well as cancer-directed treatment claims defined by biologic therapy, chemotherapy, radiation, and surgery. We used multivariable generalized linear models adjusting for demographic and clinical characteristics to examine cost differences by histologic subtype. **RESULTS:** Overall, 138,214 patients met the inclusion criteria, of whom 23.3% had squamous mNSCLC and 76.7% had nonsquamous mNSCLC. The proportion of patients receiving cancer-directed treatment was higher among those with squamous versus nonsquamous mNSCLC (72.2% vs. 60.6%; chi-square p<0.0001). Adjusted mean per-patient per-month cancer-directed treatment costs were slightly lower among patients with squamous versus nonsquamous mNSCLC (\$3,332 vs. \$3,440; p<0.0001). Additionally, adjusted mean per-patient per-month total disease-related healthcare costs were significantly lower for patients with squamous versus nonsquamous mNSCLC (\$13,862 vs. \$14,690; p<0.0001). **CONCLUSIONS:** Patients with nonsquamous mNSCLC had higher cancer-directed treatment costs and total disease-related costs compared to those with squamous mNSCLC. Given the paucity of evidence comparing costs of squamous versus nonsquamous mNSCLC, this research should be reproduced in similar settings using other real-world data sources to evaluate the generalizability of these findings.

PCN180

A POPULATION-BASED STUDY OF RADIOTHERAPY USE IN THE LAST 30 DAYS OF LIFE FOR PATIENTS WITH THORACIC CANCERS IN TAIWAN

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OBJECTIVES: Overly aggressive cancer treatment toward the end of life (EOL) is considered an indicator of poor-quality care. While radiotherapy can be used to alleviate symptoms arising from cancer, prolonged courses of radiotherapy may subject patients to additional harms. We hypothesized that among patients with thoracic cancers overly aggressive EOL radiotherapy is more likely among younger patients. **METHODS:** We defined the duration of EOL as the last 30 days of life and used data from the population-based Taiwan cancer registry linked to Taiwan death registry to construct an EOL study cohort from patients diagnosed with lung or esophageal between 2008 and 2010 and died before December 31, 2012. We identified radiotherapy use from the National Health Insurance claims data and conducted